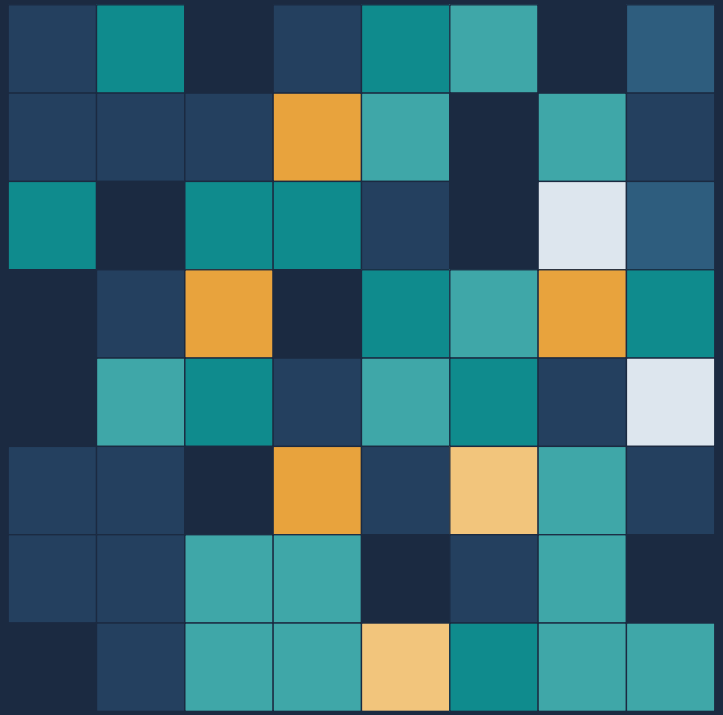
Spatial Resolution in Image Processing

ITA0522-Computer Vision

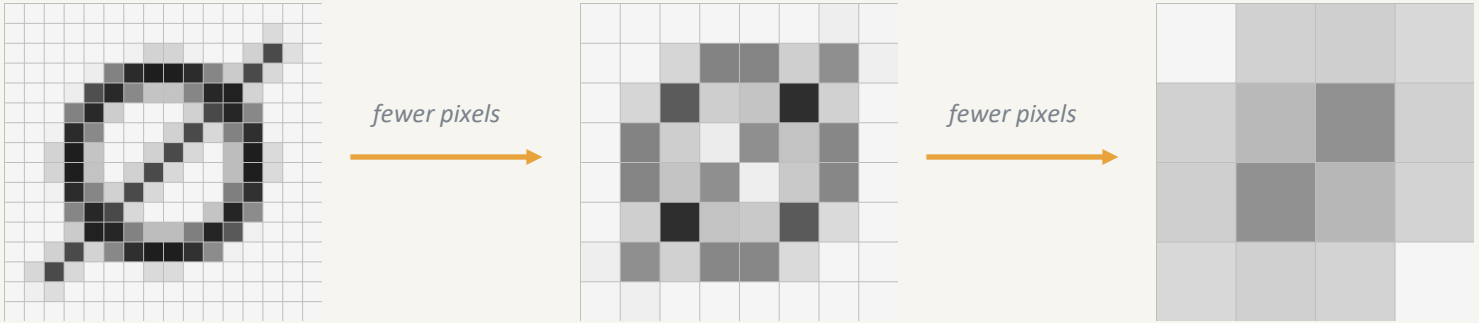
Guided by :
Dr. Meganathan E

Presented by: [M.Rohith/192311184]
[A G. Subish/192312139]



01 What Is Spatial Resolution?

- Spatial resolution means how much detail an image shows.
- An image is made of tiny squares called pixels.
- More pixels usually show more detail; fewer pixels lose detail.
- Enlarging a small image cannot fully bring lost detail back.



16 x 16 pixels **8 x 8 pixels** **4 x 4 pixels**

02 Real-World Problem: License Plate Recognition

- ANPR means Automatic Number Plate Recognition.
- It is used at toll plazas, parking areas and for traffic monitoring.
- OCR reads the letters and numbers on a detected plate.
- Low resolution can make similar characters hard to tell apart.

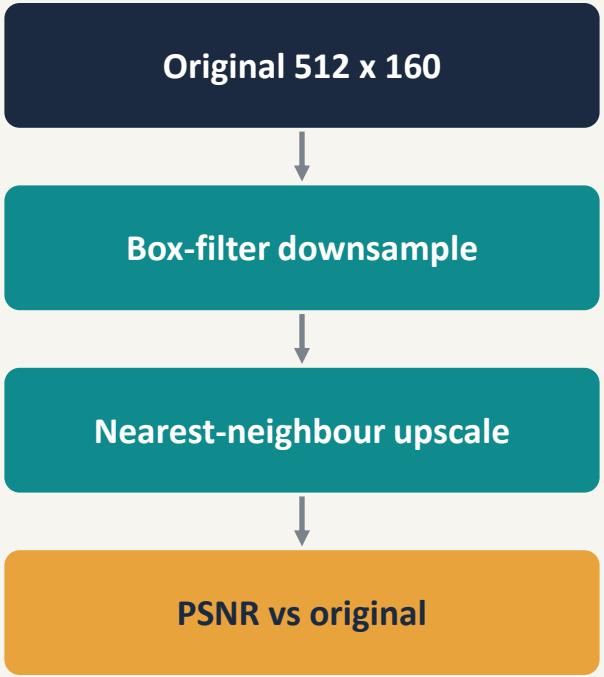
Steps in a typical ANPR system



The camera image limits how much detail later steps can use.

03 Experiment Setup

- We start with a sample plate: TN09 AB1234.
- The original image is 512 × 160 pixels.
- We reduce the width to 128, 64 and 32 pixels, then enlarge it.
- PSNR compares each result with the original image.

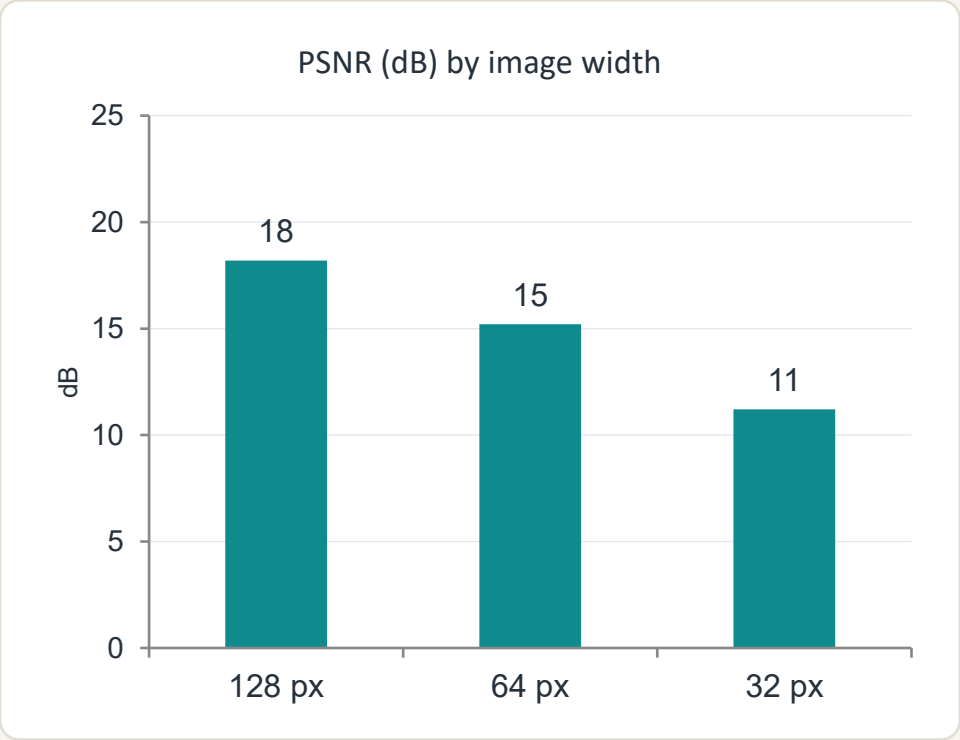


04 Visual Comparison: Resolution vs Legibility



- 512 px: characters are clear and easy to read.
- 128 px: still readable, but edges look rough.
- 64 px: some character strokes merge and look similar.
- 32 px: much of the detail is lost, so reading is difficult.

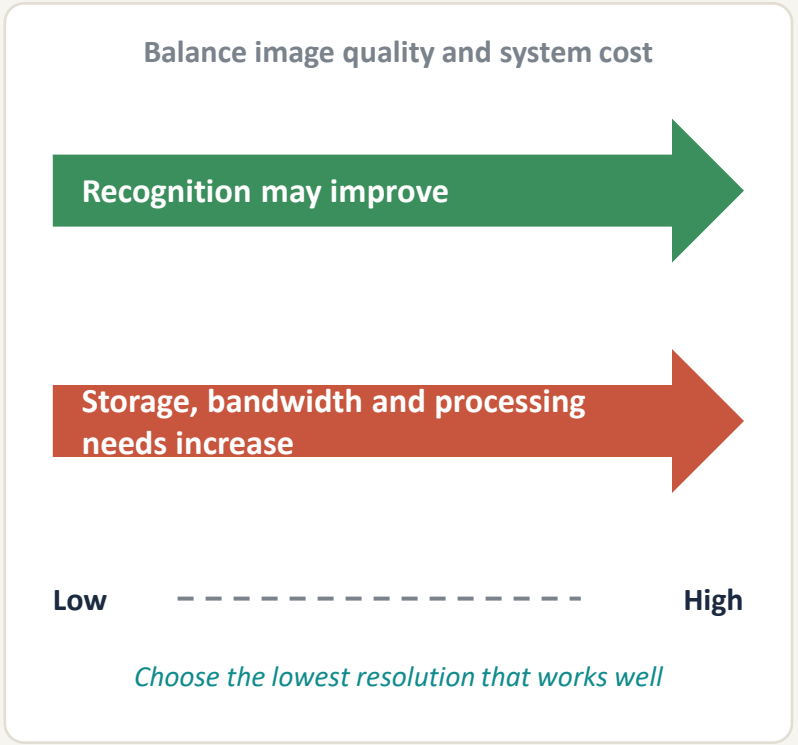
05 Quantitative Analysis: PSNR vs Resolution



- At 128 pixels, PSNR is 18.2 dB.
- At 64 pixels, PSNR is 15.2 dB.
- At 32 pixels, PSNR is 11.2 dB.
- Lower PSNR means the result differs more from the original.

06 Impact on System Performance

- Low resolution can cause more number-plate reading errors.
- Higher resolution can show more detail, but needs more storage and processing.
- A useful starting guide is about 20–30 pixels in character height.
- Use a suitable camera, zoom or closer position to capture more detail.



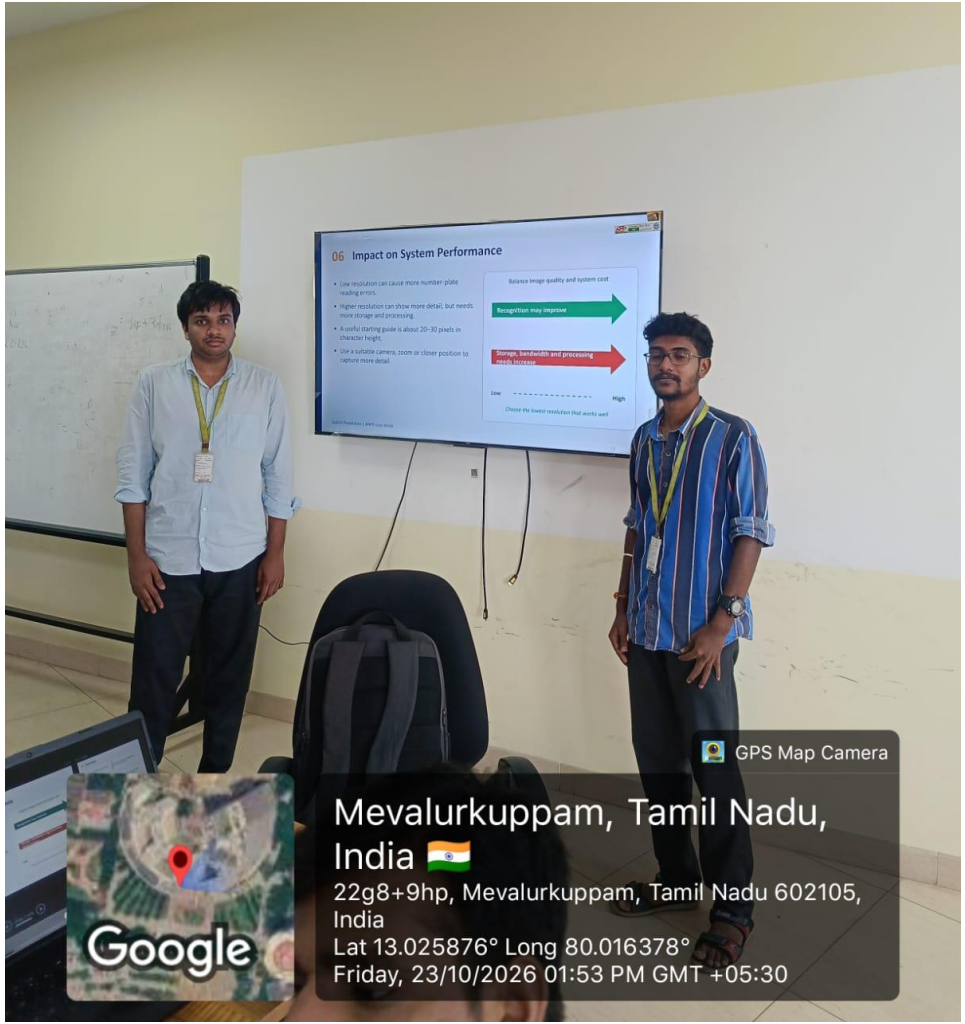
07 Conclusion

Spatial resolution affects how much detail an image contains.

In our test, 128 px remained readable; 32 px lost much of its detail.

Choose a resolution that meets the task requirements.

Enlarging cannot fully restore lost details. Good capture matters.


06 Impact on System Performance
Low resolution can cause more number-plate reading errors.
Higher resolution can show more detail, but needs more storage and processing.
A useful starting guide is about 20–30 pixels in character height.
Use a suitable camera, zoom or closer position to capture more detail.
Balance image quality and system cost
Recognition may improve
Storage, bandwidth and processing needs increase
Low
High
Choose the lowest resolution that works well
GPS Map Camera
Google
Mevalurkuppam, Tamil Nadu, India 🇮🇳
22g8+9hp, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.025876° Long 80.016378°
Friday, 23/10/2026 01:53 PM GMT +05:30

Thank You!

Spatial Resolution in Image Processing
– ANPR Case Study –



Better Resolution...
Better Recognition...

